Demonstration of Proposed Features

Date	28 October 2026
Team ID	[Insert Your Actual Team ID, e.g., TM-1042]
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	**Data Preprocessing & Cohort Stratification** Automated Data Cleaning & Cohort Feature Engineering	Validated data quality (0 missing values, 0 duplicates across 5,000 records) and binned continuous age/BMI fields into analytical tiers (Age Brackets, BMI Category).	Implemented	Yes	Verified 100% record integrity across 16 variables in Snowflake and Python ETL pipeline.
2	**Dynamic Risk Tier Classification Model** Dynamic Multi-Variable Risk Stratification	Scored all 5,000 patient records with a calculated Dynamic Risk Tier Index (BP, glucose, cholesterol, BMI, smoking) and categorized them as High Risk (score ≥ 75, 1,434 patients), Moderate Risk (50–74, 2,771) and Low Risk (< 50, 795).	Implemented	Yes	Successfully demonstrated dynamic risk tier recalculation on Executive Overview Dashboard.
3	**Interactive Clinical Risk & Vitals Dashboard** Multi-Variable Vitals Scatter Plot & Lifestyle Matrix	Rendered BMI vs. Glucose scatter plot (colored by Dynamic Risk Tier Index) with a lifestyle matrix of alcohol and smoking status, plus exercise-level vital comparisons.	Implemented	Yes	Demonstrated that smokers show a 2.2x higher High Risk share (39.3% vs. 17.7% for non-smokers); alcohol shows no added effect.
4	**Symptom Burden Co-Occurrence Heatmap** Multi-Morbidity Correlation & Resource Matrix	Visualized co-occurrence of chronic conditions (Diabetes, Heart Disease, Stroke) against symptom burden count (chest pain, fatigue, dizziness) for clinical bed/triage prioritization.	Implemented	Yes	Displayed clear risk concentration clusters guiding hospital ICU bed allocation.
5	**Guided Executive Narrative Story** 4-Scene Interactive Executive Story Progression	Built sequential story container transitioning stakeholders from macro demographic trends down to localized clinical interventions.	Implemented	Yes	Fully demonstrated smooth scene navigation across all 4 narrative points in Tableau.
6	**Embedded Web Shell & Container Shell** Flask Web Shell Integration via Tableau JS API v2	Embedded Tableau visual dashboards and story containers into a responsive HTML5/CSS3 web shell running locally on http://127.0.0.1:5001.	Implemented	Yes	Confirmed seamless web interaction, responsive layout rendering, and sub-second filter responsiveness.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%